Constitution, for example. But how can humans probe and correct a computer mythology we don't understand?

One potential guardrail is to train computers to be aware of their own fallibility. As Socrates taught, being able to say 'I don't know' is an essential step on the path to wisdom. And this is true of computer wisdom no less than of human wisdom. The first lesson that every algorithm should learn is that it might make mistakes. Baby algorithms should learn to doubt themselves, to signal uncertainty and to obey the precautionary principle. This is not impossible. Engineers are already making considerable headway in encouraging AI to express self-doubt, ask for feedback and admit its mistakes.[69]

Yet no matter how aware algorithms are of their own fallibility, we should keep humans in the loop, too. Given the pace at which AI is developing, it is simply impossible to anticipate how it will evolve and to place guardrails against all future potential hazards. This is a key difference between AI and previous existential threats like nuclear technology. The latter presented humankind with a few easily anticipated doomsday scenarios, most obviously an all-out nuclear war. This meant that it was feasible to conceptualise the danger in advance, and explore ways to mitigate it. In contrast, AI presents us with countless doomsday scenarios. Some are relatively easy to grasp, such as terrorists using AI to produce biological weapons of mass destruction. Some are more difficult to grasp, such as AI creating new psychological weapons of mass destruction. And some may be utterly beyond the human imagination, because they emanate from the calculations of an alien intelligence. To guard against a plethora of unforeseeable problems, our best bet is to create living institutions that can identify and respond to the threats as they arise.[70]

Ancient Jews and Christians were disappointed to discover that the Bible couldn't interpret itself, and reluctantly maintained human institutions to do what the technology couldn't. In the twenty-first century, we are in an almost opposite situation. We devised a technology that *can* interpret itself, but precisely for this reason we had better create human institutions to monitor it carefully.

To conclude, the new computer network will not necessarily be either bad or good. All we know for sure is that it will be alien and it will be fallible. We therefore need to build institutions that will be able to check not just familiar human weaknesses like greed and hatred but also radically alien